

Hotel Booking Revenue Analysis

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Introduction

The hospitality industry generates a vast amount of booking data that can be leveraged to improve business performance, optimize revenue strategies, and enhance customer experience. This project, **Hotel Booking Revenue Analysis**, focuses on analyzing hotel reservation data from **Resort Hotels** and **City Hotels** to uncover valuable business insights related to revenue generation, customer behavior, market segmentation, pricing performance, and operational demand. This project demonstrates practical applications of **SQL for business analytics**, including data aggregation, KPI calculation, segmentation analysis, and reporting, while showcasing the process of transforming raw booking data into actionable business insights through data visualization and analytical storytelling.

Objectives

- Analyze hotel booking performance
- Calculate revenue metrics
- Identify high-performing market segments
- Analyze parking demand
- Determine top revenue-generating countries

Dataset Description

The dataset contains hotel booking records for **Resort Hotels** and **City Hotels** during **2018**. It includes guest demographics, booking details, reservation status, length of stay, market segments, pricing information, and additional service requests. The dataset was used to analyze hotel performance, revenue generation, customer behavior, and operational demand.

Dataset Information

Attribute	Description
Dataset Name	Hotel Booking Dataset
Total Records	21,996 Bookings
Total Columns	32 Features
Hotel Types	Resort Hotel, City Hotel
Time Period	2018
Analysis Tools	SQL Server, Python, VS Code, Tableau

Category	Column Name	Description
Booking Information	hotel	Type of hotel (Resort Hotel or City Hotel)
	is_canceled	Indicates whether the booking was canceled (1 = Yes, 0 = No)
	lead_time	Number of days between booking and arrival
	arrival_date_year	Arrival year
	arrival_date_month	Arrival month
	arrival_date_week_number	Week number of arrival
	arrival_date_day_of_month	Day of arrival
Stay Information	stays_in_weekend_nights	Number of weekend nights booked
	stays_in_week_nights	Number of weekday nights booked
	adults	Number of adults
	children	Number of children
	babies	Number of babies
Customer Information	country	Customer country code
	market_segment	Booking market segment
	distribution_channel	Distribution channel used
	customer_type	Customer category
Reservation Information	reserved_room_type	Reserved room category
	assigned_room_type	Assigned room category
	booking_changes	Number of booking modifications
	deposit_type	Deposit type applied
Financial Information	adr	Average Daily Rate (average room revenue per night)
Additional Services	required_car_parking_spaces	Number of parking spaces requested
	total_of_special_requests	Number of special requests made by guests
Reservation Status	reservation_status	Booking status (Check-Out, Canceled, No-Show)
	reservation_status_date	Date of booking status update

1. Revenue by Hotel Type (2018)

Hotel Type	Revenue
Resort Hotel	3,526,408.51
City Hotel	3,291,708.05

Insight:

- Resort Hotels generated slightly more revenue than City Hotels.
- Resort Hotels contributed approximately **51.7%** of total revenue.

→ City Hotels contributed approximately **48.3%**.

2. Total Revenue

Year	Revenue
2018	6,818,116.56

Insight:

- The dataset contains only **2018 data**, so year-over-year growth cannot be analyzed.
 - Total revenue generated was **\$6.82 million**.
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3. Parking Demand

Year	Parking Demand
2018	1,352

Insight:

- 1,352 parking spaces were requested.
 - Since only one year of data is available, we cannot determine whether parking demand is increasing or decreasing.
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4. Average Daily Rate (ADR)

Year	Average ADR
2018	87.18

Insight:

- On average, guests paid **87.18 per room per night**.
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5. Revenue by Market Segment

Market Segment	Revenue
Online TA	2,465,380.36
Offline TA/TO	1,863,581.69
Groups	1,330,906.36
Direct	966,255.87
Corporate	190,955.29
Complementary	988.99
Undefined	48.00

Insight:

- **Online Travel Agencies (Online TA)** are the biggest revenue source.
 - Travel agents and tour operators contribute significantly.
 - Corporate bookings contribute a relatively small share.
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6. Top Revenue-Generating Countries

Country	Revenue
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PRT (Portugal)	3,790,948.10
ESP (Spain)	605,351.13
GBR (United Kingdom)	597,889.26
FRA (France)	376,916.74
IRL (Ireland)	234,117.54
ITA (Italy)	165,290.69
DEU (Germany)	147,909.17
BEL (Belgium)	81,868.99
NLD (Netherlands)	77,902.89
CHE (Switzerland)	64,607.54

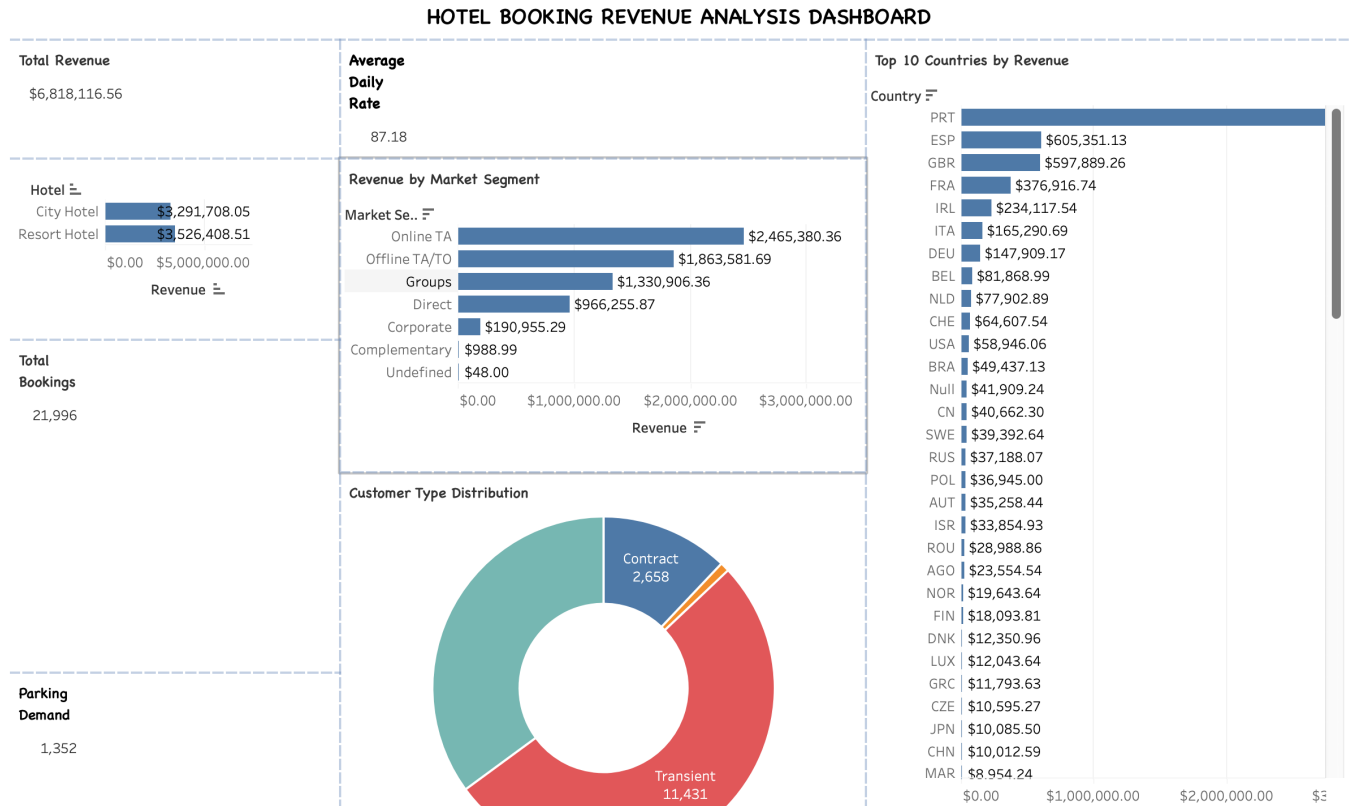
Insight:

- Portugal dominates the customer base and revenue.
 - Spain and the UK are the next most valuable markets.
 - Most revenue comes from European countries.
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7. SQL Skills Demonstrated

- SELECT
- GROUP BY
- ORDER BY
- SUM()
- AVG()
- ROUND()
- Revenue calculations
- Business analytics

8. Key Results



- Total Revenue: **\$6.82M**
- Highest Revenue Hotel Type: **Resort Hotel**
- Top Market Segment: **Online TA**
- Top Country: **Portugal**
- Average Daily Rate: **87.18**
- Parking Demand: **1,352 spaces**

9. Conclusion

The analysis reveals that **Resort Hotels** slightly outperform **City Hotels** in revenue generation, while **Online Travel Agencies (OTA)** remain the dominant booking channel. **Portugal** contributes **the highest share of revenue**, highlighting the importance of the domestic market. These insights can support strategic decisions related to pricing, marketing, customer acquisition, and operational planning.